

Disproof of conjecture 00000001737

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Verdict: FALSE

We disprove conjecture 00000001737 as stated. The refutation is unconditional and elementary: for the two-element prime set $S = \{2, 3\}$ the projective line minus three points $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ has 21 S -integral points, not at most 12. Since $21 > 12$, the conjectured maximum is false. We additionally verify that 12 is not a value attained by any two-element prime set S with $p < q \leq 23$, and that the true maximum there is 21, attained at $S = \{2, 3\}$. The final phrase of the conjecture, “with the extremum exhausted by explicit hyperelliptic covers”, is a non-sequitur: $\mathbb{P}^1$ minus three points has genus 0 and is not a hyperelliptic (nor any) curve of genus ≥ 1 ; no hyperelliptic cover is involved.

1 The conjecture, quoted verbatim

From `conjectures/00000001737.md`:

English. Conjecture: The maximal possible number of S -integral points of $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ with S of two primes is 12; a tightening of Evertse’s general bound, with the extremum exhausted by explicit hyperelliptic covers.

The conjecture is filed in both English and Chinese; the Chinese original is reproduced verbatim (in Unicode) in `README.md`, and the two versions are identical in content. The PDF font does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here.

2 The definition of S -integral point

The statement is only meaningful under the standard definition, and it is worth fixing carefully because the naive reading is degenerate.

Let $K = \mathbb{Q}$, let $S = \{p_1, p_2\}$ be a finite set of primes, and let

$$\mathcal{O}_S = \mathbb{Z}[1/p_1, 1/p_2] = \mathbb{Z}[1/p_1 p_2]$$

be the ring of S -integers. An element of $\mathcal{O}_S^\times$ is an S -unit: a non-zero rational u with $|u| = p_1^a p_2^b$ for some $a, b \in \mathbb{Z}$.

Definition 1. The scheme $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ over $\mathbb{Z}$ is $\text{Spec } \mathbb{Z}[x, 1/x, 1/(1-x)]$. An S -integral point is a $\mathbb{Z}$ -morphism $\text{Spec } \mathcal{O}_S \rightarrow \mathbb{P}^1 \setminus \{0, 1, \infty\}$, i.e. a rational x such that

$$x \in \mathcal{O}_S^\times, \quad 1-x \in \mathcal{O}_S^\times.$$

Equivalently: x , $1/x$ and $1/(1-x)$ have no prime outside S in numerator or denominator. Note that “ $1/x$ integral” forces $v_p(x) = 0$ for $p \notin S$.

The integrality is required at the *whole* divisor $\{0, 1, \infty\}$. This is the only reading under which the problem is a finite count. The naive reading “ $x \neq 0, 1$ with $x \in \mathcal{O}_S$ ” would already make the set infinite: $\mathbb{Z}[1/6]$ is infinite, and every $x \in \mathbb{Z}[1/6]$ with $x \neq 0, 1$ would qualify. So the naive reading cannot be what is meant, and we use Definition 1 throughout.

Remark 2 (classical form). Writing $y = 1 - x$, Definition 1 says precisely that (x, y) is a solution in S -units of

$$x + y = 1.$$

This is the *S -unit equation*. By Mahler’s theorem (1933) it has finitely many solutions, effective bounds going back to Györy (1979); the deep inputs (linear forms in logarithms, or lattice reduction) are needed only for *exact* computation. For the refutation we need only a lower bound, which is elementary.

3 The decisive witness: $S = \{2, 3\}$

Theorem 3. *For $S = \{2, 3\}$ the set of S -integral points of $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ has exactly 21 elements, namely*

$$\begin{aligned} & -8, -3, -2, -1, -\frac{1}{2}, -\frac{1}{3}, -\frac{1}{8}, \\ & \frac{1}{9}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{8}{9}, \frac{9}{8}, \frac{4}{3}, \frac{3}{2}, 2, 3, 4, 9. \end{aligned}$$

Verification. Each listed x has x and $1 - x$ of the form $\pm 2^a 3^b$ with $a, b \in \mathbb{Z}$; the corresponding pairs $(x, 1 - x)$ are

$$\begin{array}{cccc} (-8, 9) & (-3, 4) & (-2, 3) & (-1, 2) \\ (-\frac{1}{2}, \frac{3}{2}) & (-\frac{1}{3}, \frac{4}{3}) & (-\frac{1}{8}, \frac{9}{8}) & (\frac{1}{9}, \frac{8}{9}) \\ (\frac{1}{4}, \frac{3}{4}) & (\frac{1}{3}, \frac{2}{3}) & (\frac{1}{2}, \frac{1}{2}) & (\frac{2}{3}, \frac{1}{3}) \\ (\frac{3}{4}, \frac{1}{4}) & (\frac{8}{9}, \frac{1}{9}) & (\frac{9}{8}, -\frac{1}{8}) & (\frac{4}{3}, -\frac{1}{3}) \\ (\frac{3}{2}, -\frac{1}{2}) & (2, -1) & (3, -2) & (4, -3) \\ (9, -8) & & & \end{array}$$

Every entry of every pair is a unit of $\mathbb{Z}[1/6]$, so all 21 rationals are S -integral points by Definition 1. They are pairwise distinct, so the count is at least 21. The exact count 21 (rather than merely ≥ 21) is confirmed by exhaustive enumeration: for every exponent bound $B \geq 5$ the enumeration of $x = \pm 2^a 3^b$ with $|a|, |b| \leq B$ satisfying Definition 1 returns the same 21 elements; the saturated list is stable for all $B \in \{5, 10, 20, 40, 60, 80, 100, 120, 150\}$. See `reproduce.py` and Section 8. $\square$

Corollary 4 (the conjecture is false). *The maximal possible number of S -integral points of $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ over two-element prime sets is at least 21. In particular it is not 12, and the assertion “the maximal possible number ... is 12” is false.*

Proof. Theorem 3 gives 21 points for the admissible two-element prime set $S = \{2, 3\}$, and $21 > 12$. $\square$

4 The maximum over all two-element prime sets

To make the failure robust, we computed the exact count for every pair of primes $p < q \leq 23$ by the same saturated enumeration.

S	number of S -integral points
$\{2, 3\}$	21
$\{2, 5\}, \{2, 7\}, \{2, 17\}$	9
$\{2, 11\}, \{2, 13\}, \{2, 19\}, \{2, 23\}$	3
any $\{p, q\}$ with p, q both odd ($3 \leq p < q \leq 23$)	0

The maximum over all these sets is 21, attained only at $S = \{2, 3\}$. No two-element prime set in the range has exactly 12 points: the observed counts are 0, 3, 9, 21 only. The same is expected in general: $S = \{2, 3\}$ is the extremal two-element set for the S -unit equation.

Remark 5 (cross-reference). The values for $S = \{\text{first } n \text{ primes}\}$ are OEIS **A362567**, “Number of rational solutions to the S -unit equation $x + y = 1$, where $S = \{\text{prime}(i) : 1 \leq i \leq n\}$ ”, whose data begin

$$a(0) = 0, \quad a(1) = 3, \quad a(2) = 21, \quad a(3) = 99, \dots$$

with offset 0. Its $a(2) = 21$ is exactly our count, and its EXAMPLE section lists precisely the 21 solutions exhibited above. Our enumeration reproduces A362567’s $a(2)$ independently and from scratch.

5 Comparison with Evertse’s bound

Evertse’s general bound for the number of solutions of the S -unit equation is *exponential* in $|S|$; the commonly quoted form is

$$N(S) \leq 3 \cdot 7^{2|S|+3}.$$

For $|S| = 2$ this is

$$3 \cdot 7^7 = 3 \cdot 823543 = 2\,470\,629,$$

which is not close to 12. It is also not close to 21. So:

- 12 is not Evertse’s bound for $|S| = 2$; the true Evertse-type bound is 2 470 629.
- 12 is not Evertse’s bound for any small $|S|$ either: for $|S| = 1$ the bound is $3 \cdot 7^5 = 50409$ (and the true value is $a(1) = 3$), and for $|S| \geq 2$ the bound is already in the millions. In fact 12 is not a standard bound in this circle of ideas at all.
- Therefore “a tightening of Evertse’s general bound” to 12 cannot be obtained by any known bound: the claim is false on both counts, as a maximum (it is 21 for $|S| = 2$) and as an upper bound (Evertse’s is exponential, not 12).

The genuine exact values ($a(1) = 3$, $a(2) = 21$, $a(3) = 99$) are far below Evertse’s bound; that gap is exactly why exact S -unit computations are hard, and it is unrelated to the number 12.

6 The “hyperelliptic covers” clause

The conjecture ends: “with the extremum exhausted by explicit hyperelliptic covers”. This is a non-sequitur for the objects at hand.

- $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ is a curve of genus 0: the projective line has genus 0 and removing points does not change the genus. Genus-0 curves are *not* hyperelliptic in the usual sense (no degree-2 map to $\mathbb{P}^1$ of the required kind exists over $\mathbb{Q}$ with the relevant ramification — the only genus-0 case is the base itself). There is no hyperelliptic curve and no cover to speak of.
- The S -unit equation $x + y = 1$ is a problem on the torus $\mathbb{G}_m^2$, not on a hyperelliptic curve; its finiteness comes from Mahler’s theorem and its effective resolution from linear forms in logarithms, not from hyperelliptic covering arguments.
- The extremal set for $|S| = 2$ is $S = \{2, 3\}$, with 21 points found by direct enumeration of S -units; no hyperelliptic cover is needed or used.

We therefore record the clause as mathematically unrelated to the claim, and note that it does not rescue the conjecture.

7 Framing: this is a lower-bound witness

We state the logical shape of the refutation precisely so as not to overclaim.

- We prove *existence* of 21 pairwise-distinct S -integral points for $S = \{2, 3\}$. Hence the maximum over two-element prime sets satisfies $\max \geq 21$.
- Since $21 > 12$, the conjectured maximum of 12 is refuted. This is all that is needed; an upper bound, or an exact maximum over all S , is not required.
- We do *not* claim to compute the exact maximum over all two-element prime sets. We do show that for $S = \{2, 3\}$ the count is exactly 21 by saturation of the enumeration, and for $p < q \leq 23$ we list the exact counts, whose maximum is 21. A proof that no larger set S does better would require the S -unit theorem / linear forms in logarithms and is beyond this elementary witness.
- Likewise we do not compute exact values for general $|S|$; those are the deep computations of A362567.

8 Reproducibility

The exhaustive check is carried out by `reproduce.py` using the Python 3 standard library only (no `sympy`), with exact arithmetic via `fractions.Fraction`:

```
$ python3 reproduce.py
... (enumeration of  $\pm 2^a 3^b$  with exponent bound 60;
the full 21-element list; saturation across bounds 5, ..., 150;
counts for all prime pairs  $p < q \leq 23$ ; the checks)
PASS: 21 distinct S-integral points exist for  $S = \{2, 3\}$ ;
21 > 12, so conjecture 00000001737 is FALSE.
```

The refutation is also formalised in Lean 4 under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). The main statements are

```
theorem isSmooth_sound {n : Nat} (h : IsSmooth n = true) :
  ∃ a b : Nat, n = 2 ^ a * 3 ^ b
theorem sIntegral_sound {p : Int} {q : Nat} (h : SIntegral p q) :
  0 < q ∧ p ≠ 0 ∧ (∃ a b, p.natAbs = 2^a*3^b)
```

```

      ∧ (∃ a b, q = 2^a*3^b) ∧ p ≠ q ∧
      (∃ a b, ((q:ℤ)-p).natAbs = 2^a*3^b)
theorem conjecture_00000001737_false :
  pts.length = 21 ∧ (∀ x ∈ pts, SIntegral x.1 x.2) ∧
  (∀ x ∈ pts, ∀ y ∈ pts, x ≠ y →
    x.1*(y.2:ℤ) ≠ y.1*(x.2:ℤ)) ∧
  (12 : ℕ) < pts.length

```

The rationals are encoded as pairs $(p : \text{Int}, q : \text{Nat})$ (lowest terms) because `decide` cannot reduce `Rat` (its normalisation is opaque); comparisons use cross-multiplication. `sIntegral_sound` is the faithfulness bridge from the executable Boolean test to the mathematical definition. See `lean4/README.md` for the statement table, the proof strategy, and the axiom audit (which reports only `propext` and `Quot.sound`, and no `sorryAx`).

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic main.tex`.
- **PDF** — at `build/main.pdf`, produced by `tectonic --outdir build main.tex`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (library `Main`, no dependencies), `Main.lean`, `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0 and the audit reports no `sorryAx`.